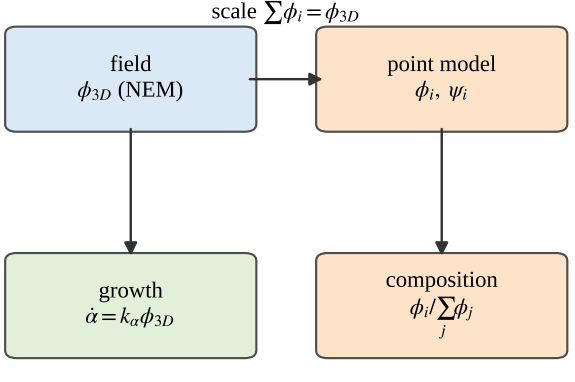
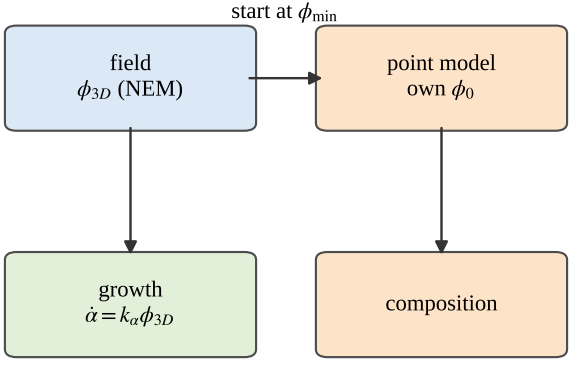
(a) scaled: amount from the field



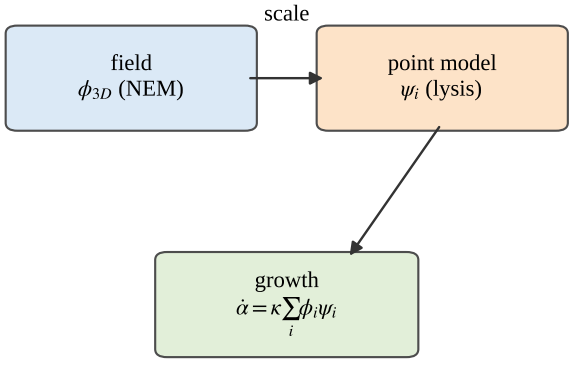
$\phi_0 = 1 - \phi_{3D}$: physical void

(b) unscaled: ϕ_0 stays auxiliary



composition depends on the local age

(c) growth from the living biomass



lower ψ_i lowers growth from inside